at the second and third levels. This is where open standards come in.

How are open standards useful?

There are five key reasons why IoT devices don't communicate with one another beyond a point.

Excessive radios and networks: A smart home may need to simultaneously leverage Wi-Fi, Zigbee, Z-Wave and Bluetooth Low Energy (BLE). LoRaWAN or NB-IoT can be implemented in a farm. However, all have different methods for transporting data.

Many data formats: One of the devices may be a JSON sender. Another one may send raw bytes. The third one may have a vendor-specific binary format. Even when two devices send JSON, the names of the fields are frequently different.

Closed vendor clouds: There are numerous vendors that wish to trap customers in their app and cloud. A device can only be used with that vendor's platform. It is difficult or impossible to move data out.

Old industrial systems: Machines used in factories may be more than 20 or 30 years old. They communicate using older protocols such as Modbus or vendor-specific fieldbus protocols. These machines will not be replaced soon.

Small, low-power devices: Numerous sensors operate on battery power for a number of years. They don't have much RAM and/or CPU. They are not intended to run heavy protocols that are created for servers.

This incompatibility can cost a lot of money. Each new device will require specific code for each app. You can have up to three apps and four device types, which can require up to 12 custom links. All links need to be written, tested and maintained. The difference an open standard can make is illustrated in Figure 1.

You can see that each device and each app connects only once to a common standard in the middle. Four devices and three apps lead to seven connections instead of 12. The bigger the system is, the more it saves.

An open standard is a technical rule book that can be read and used by everyone. It's not owned by any company. As it is written and maintained by a neutral body, e.g., OASIS, IETF or IEC, anyone can develop products on it without having to pay any licence fee.

There are four ways in which open standards benefit IoT:

A shared language: Different vendor devices have the same rules, so they can send messages to each other.

No lock-in: You can change sellers if they miss a beat or raise the price. Devices remain on the same standard.

Available open source tools: Popular standards are well-developed open source code. Developers don't have to develop a new protocol.

A large community: There are lots of people testing and improving the standard. Security bugs and holes are discovered quickly.

Open source and open standards are in partnership with each other. The rules are defined in the standard. Open source code provides a ready method. For instance, MQTT is the standard and

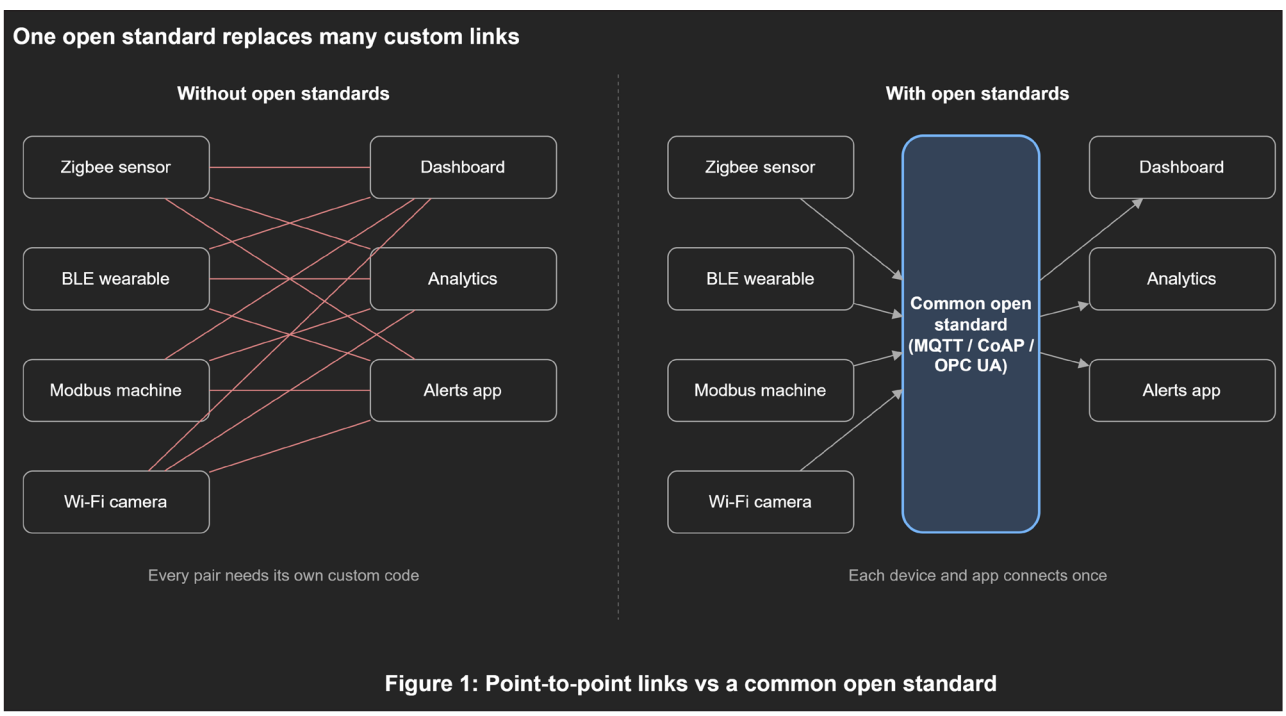


Figure 1: Open standard vs custom links